WRITTEN TEST - INSTRUCTION	
Candidate Name:	
Assessor Name:	
Qualification:	**National Skill Certificate in Web Application Development with Python, Level-4**
Units of Competency	GU-11-L4-V1 Lead Small Team SU-ICT-03-L3-V1 Comply to Ethical Standards in the ICT Workplace OU-ICT-WADP-01-L4-V1 Enable Django Framework Environment OU-ICT-WADP-02-L4-V1 Develop dynamic web pages OU-ICT-WADP-03-L4-V1 Use Django model OU-ICT-WADP-04-L4-V1 Process Forms OU- ICT-WADP-05-L4-V1 Customize UI OU-ICT-WADP-06-L4-V1 Apply User Management OU-ICT-WADP-07-L4-V1 Create API using Django REST Framework OU-ICT-WADP-08-L4-V1 Create Final project
Assessment Centre:	
Date of Assessment:	
Time Of Assessment:	

Instruction's:

Read and understand the direct Wins carefully:

- this written examination is based on the performance criteria from all the units of competency in **Web Application Development with Python, Level-4**
- this assessment activity will be used to measure your underpinning knowledge
- write your answers on the paper provided
- answer all the questions as best as possible
- you have 1 hour to complete this test

Candidate's Signature		Date:
Assessor Signature		Date:

Multiple Choice Question	
1. Which of the following is NOT a web server commonly used with Django	A. Apache **B. MySQL** C. Nginx D. IIS (Internet Information Services)
2. Which command is used to create a new Django project?	A. django new project_name B. django-admin startproject project_name C. create-django-project project_name **D. python manage.py startproject project_name**
3. In which file should you define the URL patterns for your Django application?	A. settings.py **B. urls.py** C. models.py D. views.py
4. In a typical Django project, what is the primary configuration file that contains settings for the project?	**A. settings.py** B. urls.py C. models.py D. views.py
5. What is the primary function of the views.py file in a Django project?	A. It defines the database schema and models. B. It specifies the URL patterns and their corresponding views. C. It contains the project's settings. **D. It contains the logic for handling HTTP requests and rendering templates.**
6. How can you capture a variable value form a URL in Django?	**A. Using angle brackets in the URL pattern** B. Using parentheses in the URL pattern C. Using square brackets in the URL D. Using double curly braces in the URL pattern
7. Which filter is used to join elements of a list into a single string in Django templates?	A. merge **B. join** C. concat D. combine
8. Which command is used to create the first project in Django?	**A. django-admin startproject project_name** B. django-admin startapp project_name C. django-admin create project_name D. django-admin createproject project_name
9. What is the purpose of the 'related_name' attribute in a Django model's foreign key field?	A. It specifies the name of the related model **B. I provides a way to access related objects from the related model** C. It specifies the database table name for the related model D. It defines the default value for the foreign key field
10. In Django model managers, what is the purpose of the 'create()' method?	A. Updates an existing record **B. Creates a new record in the database** C. Deletes records based on a filter D. Retrieves a single record from the database
11. Which method is commonly used in Django views to handle the Delete operation (removing data) for a specific model instance?	A. remove() B. erase() C. destroy() **D. delete()**
12. What is the purpose of 'DateTimeField' in Django models?	A. Stores only date information B. Stores only time information **C. Stores both date and time information** D. Stores duration or time interval
13. Which template tag is used to display the URL for a named URL pattern in Django templates?	**A. {% url %}** B. {% link %}

	C. (% href %} D. {% route %}
14. How do you access dictionary values in Django templates?	**A. {{ dictionary.key }}** B. {{ dictionary[key]}} C. {{ dictionary.value }} D. {{ dictionary.item }}
15. Which template tag is used for including template comments that won't be rendered in the final HTML output?	**A. {% comment %}** B. {# comment #} C. <!-- comment -- > D. {% ignore %}
16. How do you create a dynamic webpage using Django templates?	A. By writing JavaScript code **B. By defining dynamic content using template tags and filters** C. By using HTML only without any dynamic elements D) By including external PHP files
17. What is the purpose of the python manage.py migrate command in Django?	**A) It applies all the pending database migrations to the database** B) It creates new database migrations based on the changes detected in the models. C) It initializes a new Django project with a superuser account. D) It generates a new migration file for the database schema.
18. Which of the following statements is true about virtual environments in Python?	**A) Virtual environments are used to run Python code without installing Python globally on the system.** B) Virtual environments are used to create separate instances of the operating system. C) Virtual environments are not necessary in Python as it supports global package installations. D) Virtual environments can only be created using third-party packages, not with the built-in Python tools.
19. What is the purpose of the pip freeze command in Python?	A) It installs Python packages. **B) It lists all installed packages and their versions.** C) It upgrades the Python interpreter. D) It uninstalls Python packages.
20. Which of the following is used to define a primary key in a Django model?	A) PrimaryKey **B) primary_key=True** C) primary=True D) pk=True

21. What is the purpose of the migrations directory in a Django project?

Answer: The migrations directory stores files that track and apply database schema changes to keep the database in sync with your Django models.

22. What is the significance of the `name` parameter in URL pattern?

Answer: The name parameter in a URL pattern allows you to refer to that URL by name instead of hardcoding the path, making it easier to manage and change URLs in templates or views.

```
# urls.py
path('about_us/', about, name='about_us')

# In template
<a href="{% url 'about_us' %}">About Us</a>
```

23. What is the advantage of using named URL patterns in Django?

Answer: Named URL patterns make it easier to manage links in your project — you can change the URL path in one place without updating all templates or views.


```
# urls.py
path('about_us/', about, name='about_us')

# In template
<a href="{% url 'about_us' %}">About Us</a>
```


24. What does the yesno filter do in Django templates?

Answer: In Django templates, the yesno filter is used to display one of two specified strings based on whether a given value is true or false

```
{{ is_active|yesno:"Yes,No,Maybe" }}
```

If is_active is True, it shows “Yes”; if False, “No”; and if None, “Maybe.”

25. What is the purpose of the save() method in Django models?

Answer: The save() method in Django models is used to store or update an object’s data in the database.

```
student = Student(name="Rohim", age=22)
student.save()
```

26. What is the role of the 'TemplateView' class in Django CBVs?

Answer: The TemplateView class in Django CBVs is used to render a template without needing to write a custom view function. It's ideal for static pages.

27. What is the significance of the 'request.POST' dictionary in Django form submissions?

Answer: The request.POST dictionary in Django is used to access data submitted via an HTML form using the POST method. It contains all the form input fields as key-value pairs, where the key is the name attribute of the input field, and the value is the data entered by the user.

```
def contact_view(request):
    if request.method == "POST":
        name = request.POST.get('name')
        email = request.POST.get('email')
```

28. What challenges can arise when managing versioning and caching for static files in large web applications?

Answer: When managing versioning and caching for static files in large web applications, several challenges can arise:

1. Cache Invalidation: Updating static files without forcing users to reload can be difficult since browsers may use cached versions.
2. File Versioning: Ensuring each new release has unique file names or hashes to distinguish from older versions.
3. Deployment Synchronization: Serving updated files before cache clears may cause mismatches between old and new assets.
4. CDN Propagation: Cached files on CDNs may take time to update globally.

29. What are Bootstrap cards, and how are they used in web development?

Answer: Bootstrap cards are flexible and responsive content containers used to display information in a structured, stylish box. They can include text, images, buttons, and links.

card.html

```html
<div class="card" style="width: 18rem;">
  <img src="image.jpg" class="card-img-top" alt="...">
  <div class="card-body">
    <h5 class="card-title">Card Title</h5>
    <p class="card-text">This is a Bootstrap card example.</p>
    <a href="#" class="btn btn-primary">Read More</a>
  </div>
</div>
```

30. How can you authenticate users in DRF APIs?

Answer: In Django REST Framework (DRF), users can be authenticated using built-in or custom authentication classes.

Common methods:

1. Session Authentication – Uses Django's built-in session framework (mainly for web clients).
2. Token Authentication – Each user gets a unique token for API access.
3. JWT (JSON Web Token) Authentication – Uses encoded tokens for stateless, secure authentication (commonly with djangorestframework-simplejwt).
4. Custom Authentication – You can create your own authentication classes for specific logic.

```python
from rest_framework.authentication import TokenAuthentication
from rest_framework.permissions import IsAuthenticated

class UserView(APIView):
    authentication_classes = [TokenAuthentication]
    permission_classes = [IsAuthenticated]

    def get(self, request):
        return Response({"message": "Authenticated user"})
```